



DIASAFE®*plus*

Fresenius Polysulfone® dialysis fluid filter

The dialysis fluid filter

DIASAFE[®] *plus*

The quality and purity of the dialysis fluid are of major concern in modern-day renal replacement therapies, because large volumes can come into contact with the patient's bloodstream during each treatment. Endotoxins, products of bacterial lysis, present in contaminated dialysis fluid may elicit acute reactions and influence long-term complications in patients on chronic hemodialysis¹.

The international standard ISO 23500-3 (2019) contains minimum requirements on endotoxin levels and on microbial contaminations for the water to be used for preparation of dialysis fluids. In addition, the international standard ISO 23500-5 (2019) specifies the minimum quality requirements for ultrapure dialysis fluid²³.

By the application of special filters that are highly efficient in retaining bacterial contaminations, the required purity grades of ultrapure dialysis fluid can be achieved conveniently³.

The **DIASAFE *plus*** filter, located at the end of the water treatment chain, enables the production of ultrapure dialysis fluid. This is attributed to the high endotoxin retention capabilities of its Fresenius Polysulfone[®] fibers⁴.

DIASAFE *plus* is an integral part of contemporary dialysis machines. Only three handling steps are necessary to install

DIASAFE *plus* (Fig. 1):

Step 1

Open the locking levers
of the filter holder



Step 2

Slide DIASAFE plus filter
into the guide slots



Step 3

Close the locking levers –
DIASAFE plus is ready to use



Fig. 1: DIASAFE plus can be connected with three rapid handling steps only. The DIAFIX lock system ensures a safe and hygienic connection. The use of DIASAFE plus is a key step towards Good Dialysis Practice².

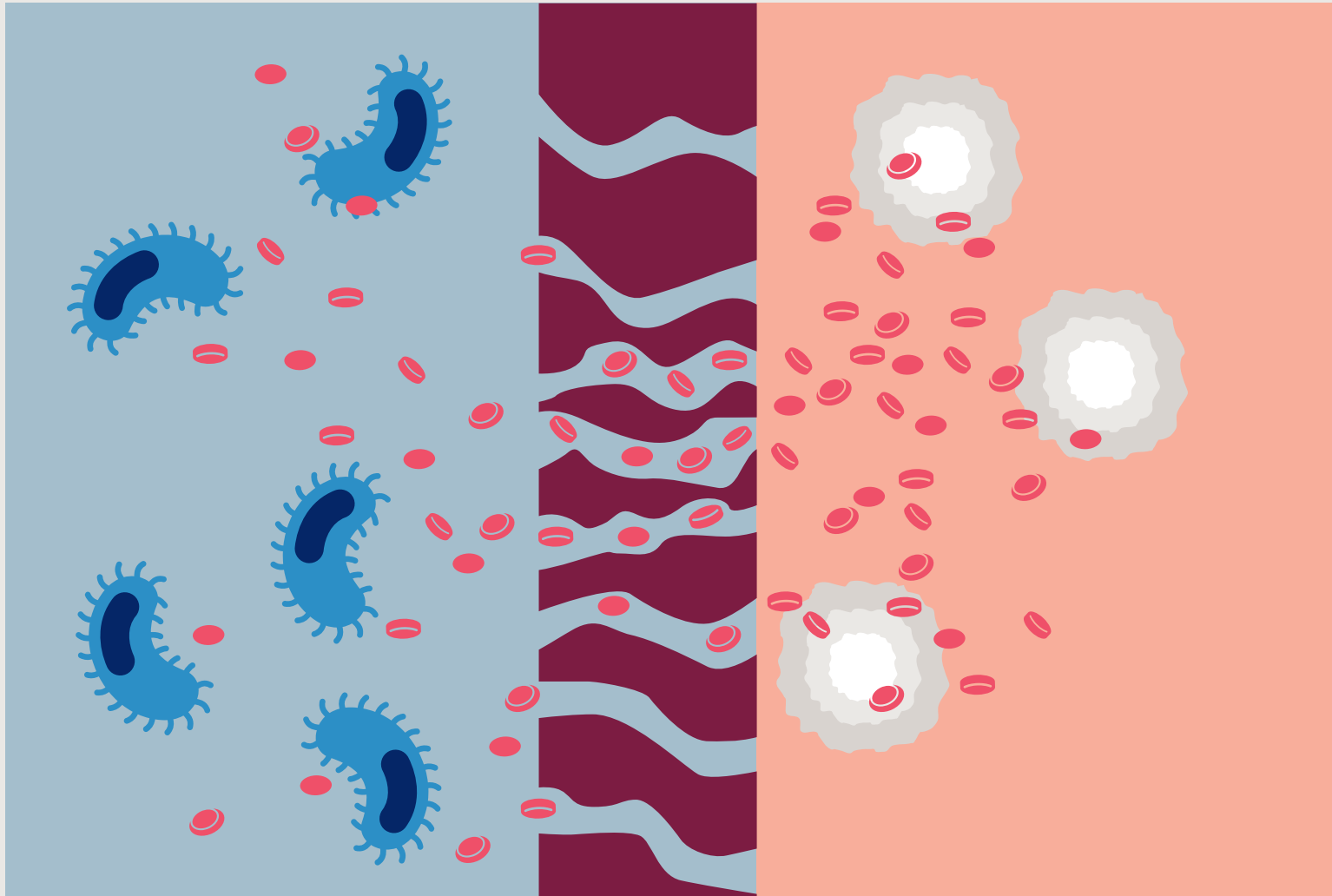
Dialysis fluid **purity**

Dialysis fluids may contain microbial impurities such as endotoxins derived from bacterial fragments. Endotoxins are known to cause acute adverse reactions and promote long-term complications in hemodialysis patients^{1,5}. The toxic properties of endotoxins can be ascribed mainly to their lipid A component, which is not exposed by intact bacteria but released only during growth or lysis of gram-negative bacteria². Endotoxin fragments may have molecular weights as small as 2000 Da². These fragments are small enough to pass across both low- and high-flux hemodialysis membranes into the patient's bloodstream (Fig. 2). With respect to permeability for bacterial products with cytokine-inducing activity, significant differences exist

between the various types of dialysis membranes, thereby offering variable degrees of safety during hemodialysis⁶. To prevent severe clinical side effects due to water contamination, the water used for hemodialysis must be purified⁷. The European Best Practice Guidelines for Hemodialysis (EBPG)⁷ advise the usage of water having a purity level in compliance with the recommendations of the European Pharmacopoeia. However, the usage of ultrapure water for conventional and high-flux dialysis is strongly recommended by the EBPG (Table 1). Ultrapure water or dialysis fluid can easily be achieved through the application of special dialysis fluid filters³.

	Regular water	Ultrapure water
Microbial contaminations (CFU/mL)	< 100	< 0.1
Bacterial endotoxins (IU/mL)	< 0.25	< 0.03

Table 1: The different purity levels of water adapted according to the EBPG⁷.



Dialysate side

Membrane

Blood side

Fig. 2: Schematic showing fragments of bacterial endotoxins which may enter the patient's bloodstream and may activate leukocytes, thereby leading to acute and chronic complications in hemodialysis patients.

Clinical advantages of using ultrapure dialysis fluid

Endotoxins can activate immune-competent cells in a number of ways, thereby contributing to chronic inflammation that is present in all hemodialysis patients⁹ (Fig. 3). Evidence demonstrates that chronic inflammation is a major risk factor for progressive atherosclerotic cardiovascular disease (CVD)⁹. Besides the application of hemodialysis membranes with a high biocompatibility, the usage of ultrapure dialysis fluid, in particular, has been shown to reduce markers of chronic inflammation in hemodialysis patients¹⁰. Therefore, it is suggestive that ultrapure dialysis fluid has a beneficial effect on inflammatory diseases such as atherosclerotic CVD¹¹. Moreover, oxidative stress – a situation in which the normal balance between production of reactive oxygen species (ROS) and antioxidant activity is tilted in favor of ROS – is increased by several treatment-related stimuli, including bacterial endotoxins derived from the dialysis fluid^{12,13}.

As oxidative stress is associated with the progression of malnutrition, anemia and inflammatory diseases such as atherosclerosis, it appears desirable to reduce dialysis-induced oxidative mechanisms, e.g. through the usage of biocompatible membranes and ultrapure dialysis fluid^{12,13}. The importance of ultrapure dialysis fluid in routine hemodialysis treatments is emphasized by the finding that endotoxins act in synergy with advanced glycation end products (AGE), which enhance inflammation and oxidative stress¹⁴. Furthermore, the use of ultrapure dialysis fluid has been shown to reduce the plasma levels of the AGE compound pentosidine¹⁵. Finally, ultrapure dialysis fluid has also been shown to improve iron utilization and the response to erythropoietin; thus, ultrapure dialysis fluid could be beneficial for anemia treatment, allowing for a reduction in erythropoietin dosage while maintaining optimal hemoglobin levels^{16,17}.

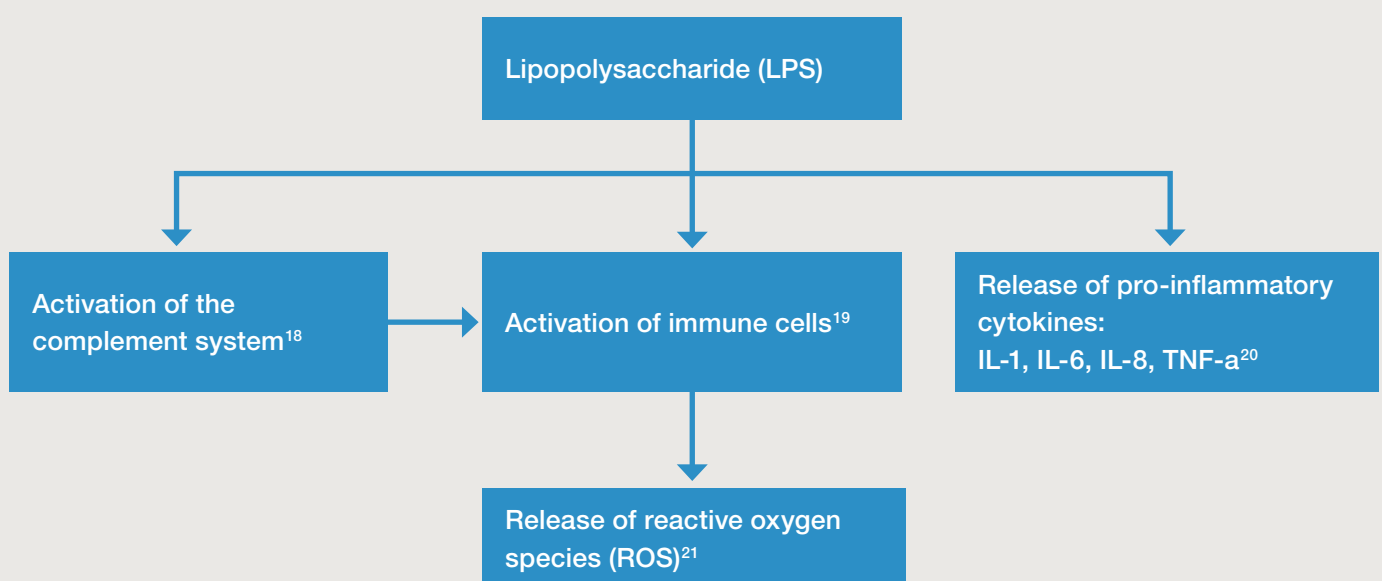


Fig. 3: Schematic showing the association between endotoxins (lipopolysaccharides), the stimulation of the release of pro-inflammatory cytokines and of reactive oxygen species.

DIASAFE[®] *plus* in the ONLINE*plus* System

Ultrapure dialysis fluid prepared with the **DIASAFE *plus* dialysis fluid filter**, together with hemodialyzers containing endotoxin-retaining membranes² (Fresenius Polysulfone[®] or Helixone[®]) are our components for a high-quality hemodialysis treatment.

The ONLINE*plus* system takes the quality standards of convective treatment modalities as hemodiafiltration / hemofiltration (HDF / HF) one step further: using two **DIASAFE *plus* dialysis fluid filters** in series, a high

level of microbiological safety can be achieved by double filtration of the substitution fluids used in ONLINE HDF / HF therapies²².

Besides a hygienic connection and ultrapure water for conventional and sterile pyrogen-free substitution fluid for convective therapy modalities, the ONLINE*plus* option also offers ONLINE priming, ONLINE bolus and ONLINE reinfusion and adds to ease of handling.

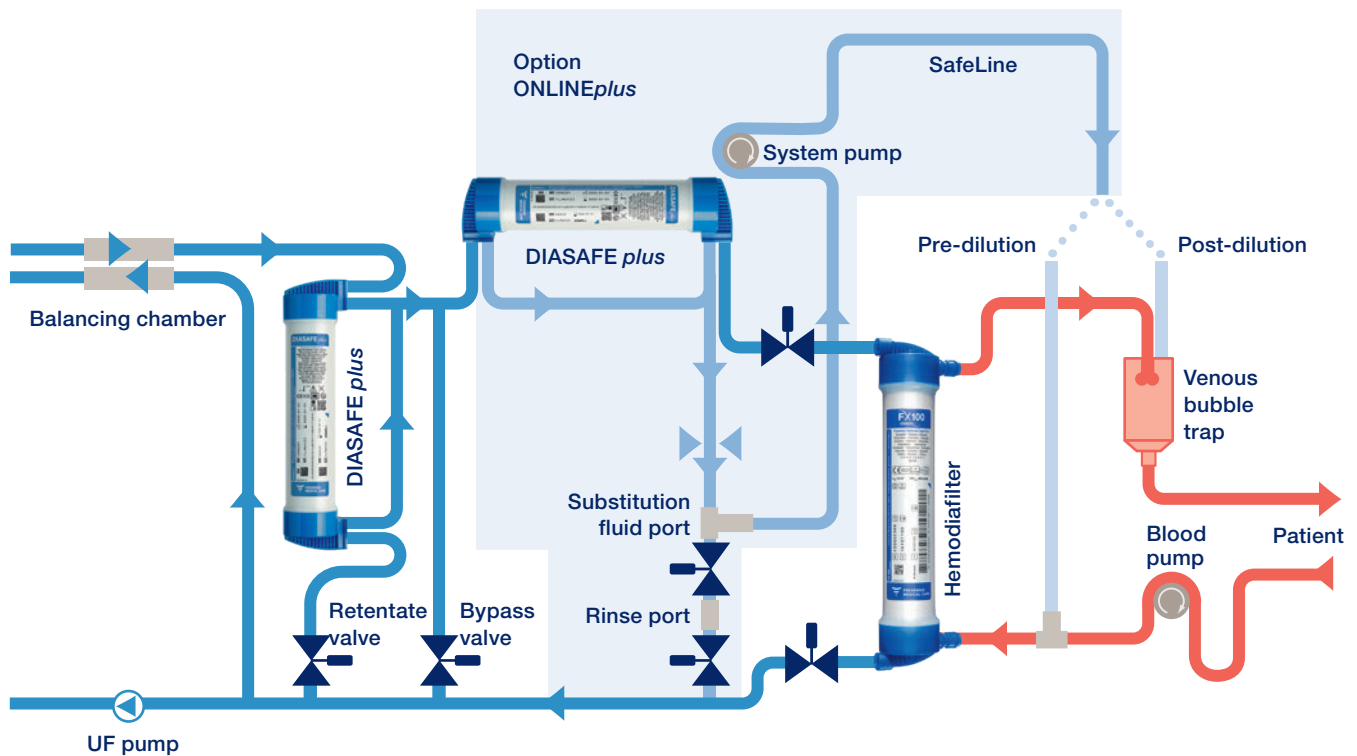


Fig. 4: Schematic flow chart of ONLINE hemodiafiltration with the ONLINE*plus* system

Technical data	Description	Article no. 500 820 1
Membrane material	Fresenius Polysulfone®	
Weight (g)	170	
Housing material	Polypropylene	
Potting material	Polyurethane	
Sealings	Silicone	
Connection to machine	DIAFIX Lock System	
Filtration rate	≥ 300 mL / (h*mmHg) (≥ 3.75 L / (min*bar); max. 2 bar)	
Operating time	Standard HD: max. 12 weeks ONLINE HDF / HF, ONLINE priming / rinsing: max. 12 weeks or 100 treatments	
Disinfection	Puristeril® 340 or Puristeril® plus (peracetic acid) Citrosteril® (citric acid) Sporotal® 100 (sodium hypochlorite) max. 11 times	

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